The Large Hadron Collider

Standard Model

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Presentation Outline

LHC: Bird's Eye view

The Proton Injection Chain

Engineering the Collisions

B Decays and LHCb

Current State of Anomaly

LHC: Bird's Eye view

- Circular tunnel, 27km in circumference (**r**=4.3 km), 100 m underground for radiation shielding.
- The ring geometry is divided into exactly **8 Points**:
  - **4 Detector Points:** Where the beams cross (ATLAS, CMS, ALICE, and **LHCb**).
  - **4 Utility Points:** Dedicated to RF acceleration, beam cleaning, and safe beam dumping.
- Protons run in two parallel beam pipes in opposite directions.

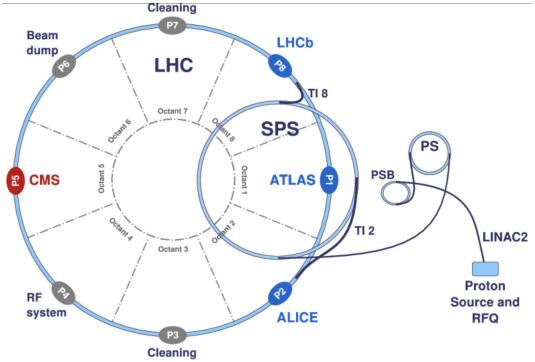


Figure: Schematic layout of the LHC ring and its detectors.

The Proton Injection Chain (Part 1): Generation & Intensity

1. Source & Linear Acceleration

- **Duoplasmatron:** Stripped hydrogen gas to provide bare protons (H^+).
  **Modern Era (Linac4): Replaced by an RF Volume Source providing H^- for charge-exchange injection.*
- **RFQ:** Uses transverse RF fields to simultaneously focus (via alternating gradients), bunch, and accelerate the DC beam.
- **LINAC2:** A Drift Tube Linac (DTL) utilizing longitudinal RF fields to accelerate bunches, shielding them inside drift tubes during the field's decelerating phase.

2. Proton Synchrotron Booster (PSB)

- **Space-Charge Mitigation:** Overcomes low-energy Coulomb repulsion limits by distributing the LINAC beam across **four vertically stacked, independent rings**.

The Proton Injection Chain (Part 2): Formatting & High Energy

3. Proton Synchrotron (PS) – The Formatter

- **Relativistic Transition (6.1 GeV):** As beam kinematics flip from classical to relativistic, the PS executes a microsecond **Phase Jump** (shifting the RF wave 180°) to maintain phase stability.
- **Bunch Splitting:** Superimposes higher-harmonic RF frequencies to deform the electrical potential "bucket," actively dividing the proton batches to create the exact **25 ns gaps** required by the LHC.

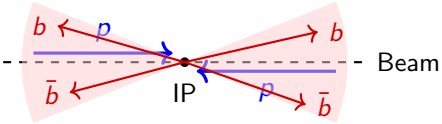
4. Super Proton Synchrotron (SPS)

- Receives the meticulously formatted, bunched 26 GeV beam from the PS.
- Accelerates this macroscopic "train" of micro-bunches up to 450 GeV for final injection into the LHC.

How the LHC Orchestrates a Collision

- **The Arc Sections (Maintaining the Beam):**
  - *Strong Focusing:* Alternating-gradient quadrupoles (FODO cells) keep the counter-rotating beams stable and roughly 1 mm wide across the 27km journey.
- **The Approach (Merging):**
  - The two parallel beam pipes merge into a single, shared vacuum chamber exactly 130 m before reaching the detector.
- **The Final Focus & Crossing:**
  - *Inner Triplet Magnets:* Three specialized quadrupoles aggressively squeeze the beam down to 16 μm (thinner than a hair) in the transverse plane.
  - *Crossing Angle:* The beams intersect at a microscopic angle ($\sim$ 300 μrad) to prevent unwanted "parasitic" collisions outside the center.

We look forward



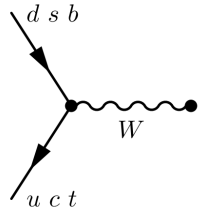
- **Production Mechanism:** Gluon-gluon fusion dominates at 13 TeV.
- **Kinematic Reality:** Heavy flavor pairs are highly correlated and strongly boosted along the beam axis.
- **Experimental Geometry:** Capturing this flux requires a dedicated, single-arm forward spectrometer, distinct from central barrel detectors.

Flavor Changing Neutral Currents (FCNCs): The GIM Mechanism

- **Charged Currents (CC): Flavor Mixing** The $W^{\pm}$ boson couples up-type to down-type quarks.

$$\mathcal{L}_{CC} \supset \frac{g}{\sqrt{2}} W_{\mu}^{+} \left(\bar{c}_L \gamma^{\mu} V_{cb} b_L\right) + \text{h.c.}$$

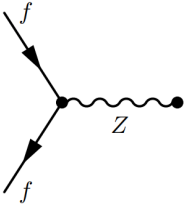
Tree-level flavor transitions ($b \to c$) are strictly allowed and have large amplitudes.



- **Neutral Currents (NC): Flavor Conserving** The Z^0 couples fields to themselves. Because the CKM matrix is unitary ($V^{\dagger} V = I$), all off-diagonal flavor-mixing terms exactly cancel out:

$$\mathcal{L}_{NC} \supset \frac{g}{2\cos\theta_W} Z_{\mu} \left(\bar{b}_L \gamma^{\mu} b_L + \bar{s}_L \gamma^{\mu} s_L\right)$$

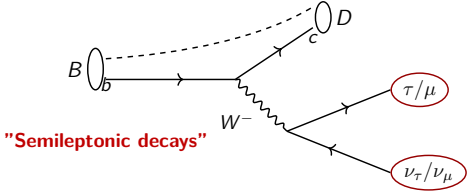
Tree-level FCNCs ($b \to s$) are mathematically forbidden. They must proceed via loops.



Flavour Anomalies: Allowed vs. Suppressed B Decays

1. Charged Current: $b \to c\ell\nu$

Tree-Level Allowed → Large decay rates

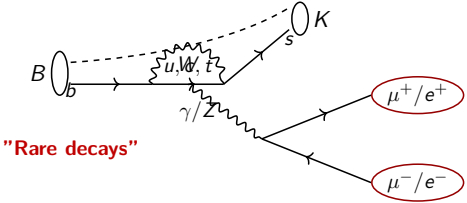


$$R_D = \frac{\mathcal{B}(B \to D\tau\nu)}{\mathcal{B}(B \to D\mu\nu)}$$

$$R_{D^*} = \frac{\mathcal{B}(B \to D^*\tau\nu)}{\mathcal{B}(B \to D^*\mu\nu)}$$

2. Neutral Current: $b \to s\ell^+\ell^-$

Loop Suppressed → Rare decays (FCNC)

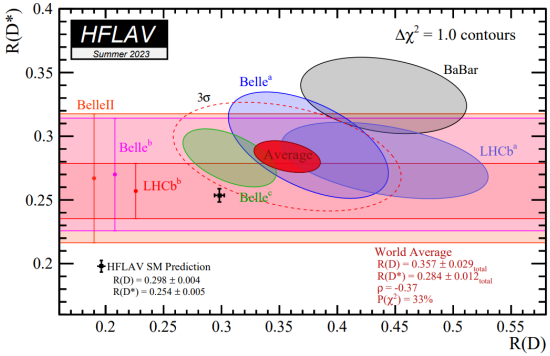


$$R_K = \frac{\mathcal{B}(B^+ \to K^+\mu^+\mu^-)}{\mathcal{B}(B^+ \to K^+e^+e^-)}$$

$$R_{K^*} = \frac{\mathcal{B}(B^0 \to K^{*0}\mu^+\mu^-)}{\mathcal{B}(B^0 \to K^{*0}e^+e^-)}$$

R_{D^*} and R_D: The 3.2σ Anomaly

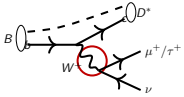
Experimental Status (HFLAV 2023)



$$R(D^*) = 0.284 \pm 0.013 \qquad R(D^0) = 0.357 \pm 0.029$$

Standard Model (Allowed CC)

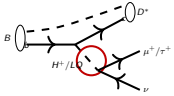
$$R_{D^*} = \frac{\mathcal{B}(B \to D^* \tau \nu)}{\mathcal{B}(B \to D^* \mu \nu)}$$



New Physics Candidates

Potential large NP effect

Involves 2nd and 3rd gen leptons.



Unpacking the Anomalies: LFU and Significance

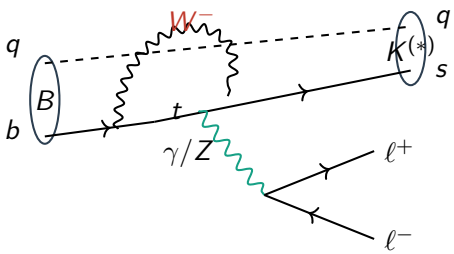
- **Why use a ratio? (Cancellation of Uncertainties)**
  - Calculating absolute branching ratios $BR(B \to D\ell\nu)$ requires knowing complex hadronic form factors (Lattice QCD).
  - By taking the ratio R_{D^*}, these massive theoretical uncertainties cleanly cancel out, resulting in an ultra-precise Standard Model prediction.

- **Lepton Flavour Universality (LFU)**
  - In the SM, the $W^{\pm}$ boson couples to e, μ, and τ with the exact same gauge coupling (g_w).
  - The only reason $R_{D^*} \neq 1$ in the SM is strictly due to phase space limits (the τ mass is much larger than the μ mass).

- **What does a 3.2σ deviation mean?**
  - The experimental world average (HFLAV) sits 3.2 standard deviations away from the SM prediction.
  - Statistically, this means there is only a $\sim 0.1\%$ chance this is a random background fluctuation. It qualifies as strong *evidence* for New Physics, though it has not yet crossed the 5σ threshold for *discovery*.

Flavour Changing Neutral Currents: $b \to s\ell^+\ell^-$

1. The FCNC Penguin Transition



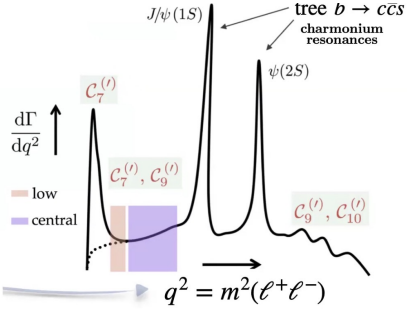
2. Effective Field Theory Framework

The heavy degrees of freedom (W, t) are integrated out into Wilson coefficients $(\mathcal{C}_i)$:

$$\mathcal{H}_{\text{eff}} \propto \sum_i \left(\mathcal{C}_i^{\text{SM}} + \mathcal{C}_i^{\text{NP}}\right) \mathcal{O}_i$$

- $\mathcal{O}_7$: Electromagnetic dipole
- $\mathcal{O}_9, \mathcal{O}_{10}$: Vector / Axial-vector

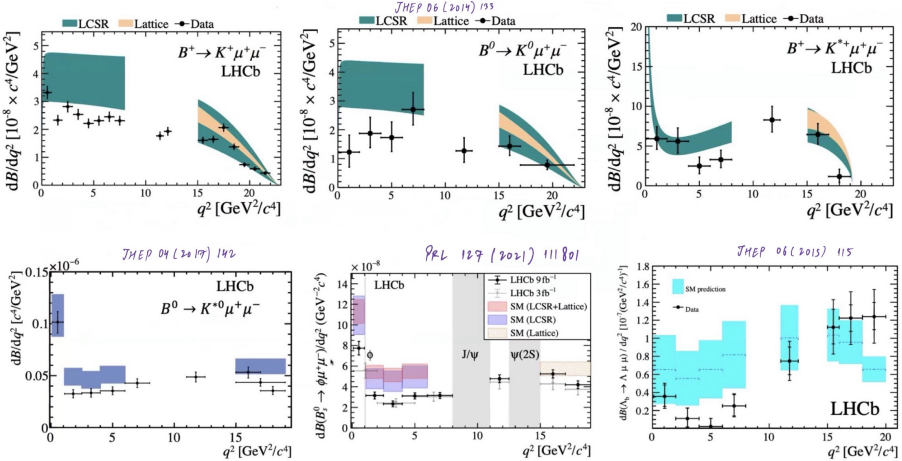
3. Differential Decay Rate Spectrum



4. Key Observables:

- **Branching Fractions:** Large hadronic uncertainties (form factors).
- **Angular (P_5'):** Cancel leading form factor dependencies.
- **Universality (R_K, R_{K^*}):** Theoretically pristine ($\sim 1\%$ SM uncertainty).

b) i. Differential decay rates of Rare Decays: $b \to s\mu^+\mu^-$



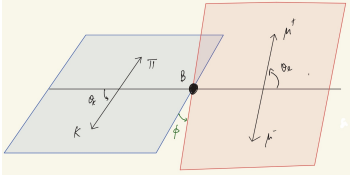
Branching fractions consistently lower than predictions.

Anomaly or theory uncertainty (20-30%)?

b) ii. Optimized Angular Observable P_5'

1. The Kinematic Challenge

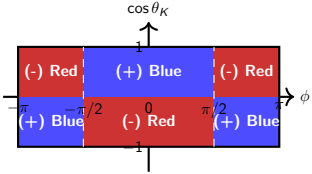
The 4-body final state is defined by three angles: $\vec{\Omega} \equiv (\cos\theta_l, \cos\theta_K, \phi)$.



- × **The Problem:** The full differential rate has **11 parameters**. Fitting this in 3D is a statistical nightmare.
- × The raw terms are infected by massive **20-30% QCD uncertainties**.

2. The Folded Asymmetry Trick

We isolate the interference term (J_5) by filtering data using its trigonometric sign: $\mathrm{sgn}(\cos\theta_K \cdot \cos\phi)$.



- ✓ **The Solution:** By counting **Blue minus Red**, 10 background parameters mathematically vanish. We cleanly extract J_5.

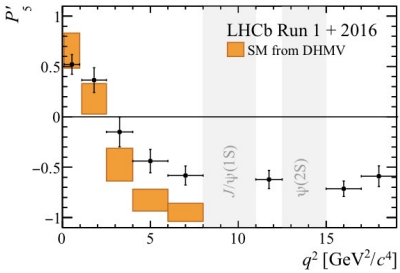
b) ii. The P_5' Anomaly: Theory Meets Data

Theoretical Payoff: Why the P_5' ratio?

$$P_5' \propto \frac{J_5}{\sqrt{-J_{2s}J_{2c}}}$$

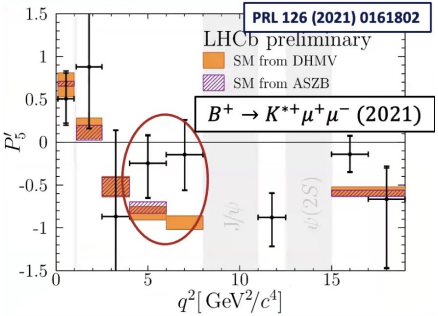
- **Cancellation:** $\sim 30\%$ QCD uncertainties mathematically cancel, yielding pristine SM predictions.
- **Target Locked:** J_5 isolates polarization interference, highly sensitive to the $\mathcal{C}_9$ **vector current**.

$B^0 \to K^{*0}\mu^+\mu^-$



[PRL 125 (2020) 011802]

$B^+ \to K^{*+}\mu^+\mu^-$



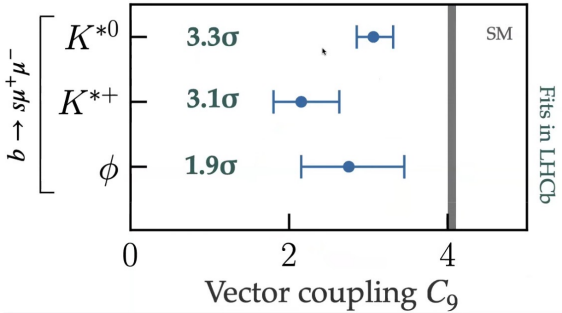
The Anomaly: Systematic deviation from the SM in the $4 < q^2 < 8$ GeV^2 bins across independent channels.

b) ii. The Global Fit: Extracting the Vector Coupling $\mathcal{C}_9$

$B^0 \to K^{*0}\mu^+\mu^-$ with 6 fb^{-1} (~4600 evts)
PRL 125 (2020) 011802

$B^+ \to K^{*+}\mu^+\mu^-$ with 9 fb^{-1} (~700 evts)
PRL 126 (2021) 161802

$B_s \to \phi\mu^+\mu^-$ with 9 fb^{-1} (~1900 evts)
JHEP 11 (2021) 043



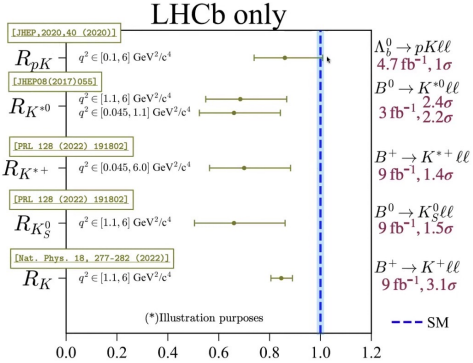
The Phenomenological Verdict

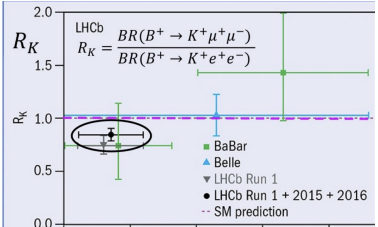
- **Statistical Authority**
  Reconstructing thousands of rare events (∼4600 in the B^0 channel) drastically shrinks the $1/\sqrt{N}$ Poisson error. High-precision deviations, reaching $\sim 3\sigma$.
- **The $\mathcal{C}_9$ Deficit**
  Deviation from standard model prediction by $\Delta\mathcal{C}_9 \approx -1$.
- No matter the spectator, short-distance coupling shifts the same. Strongly disfavors localized QCD form-factor errors.

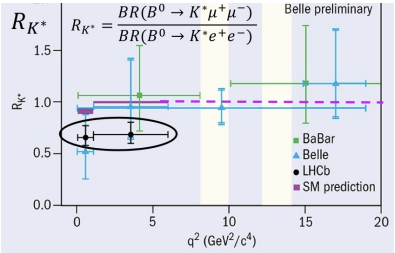
b) iii. Universality: Extracting R_K and R_{K^*}

$$R_K = \frac{\mathcal{B}(B^+ \to K^+\mu^+\mu^-)}{\mathcal{B}(B^+ \to K^+e^+e^-)}$$

$$R_{K^*} = \frac{\mathcal{B}(B^0 \to K^{*0}\mu^+\mu^-)}{\mathcal{B}(B^0 \to K^{*0}e^+e^-)}$$







b) iv. March 2022: Anomaly

The Phenomenological Convergence (March 2022)

- **Angular Geometry (P_5'):** Systematic phase-space deviations dictate a destructive shift in the vector current ($\Delta\mathcal{C}_9 \approx -1$).
- **Universality (R_K, R_{K^*}):** Theoretically pristine rate ratios independently confirm the exact same muon deficit at $> 3\sigma$ significance.

The Theoretical Conclusion:

The data overwhelmingly favors Heavy New Physics (e.g., Z' gauge bosons or Leptoquarks) coupling preferentially to the second generation.

Before declaring the Standard Model broken, LHCb had to ask one final question:

Did we truly understand the electron backgrounds?

b) iv. December 2022: The Anomaly Vanishes

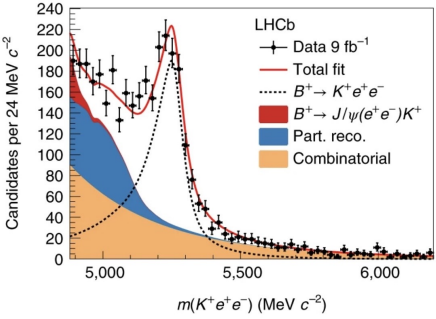
The Twist: The electrons were hiding hadronic noise.

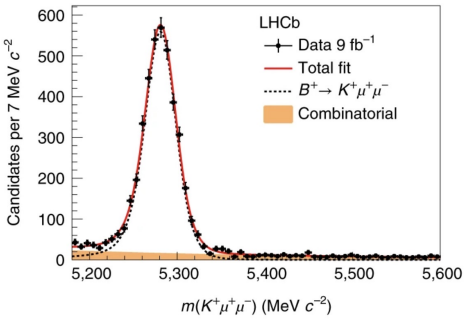
Background Mismodeling

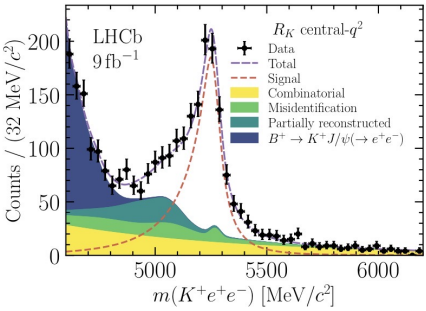
- **The Bremsstrahlung Mask:** The wide electron mass tail allowed severe background leakage to hide entirely within the signal region.
- **Misidentified Hadrons:** Strict re-analysis found that kaons and pions were occasionally faking electron signatures in the detector.
- **The Math Correction:** When these fakes were properly subtracted out, the true electron signal yield (N_{ee}) dropped significantly.

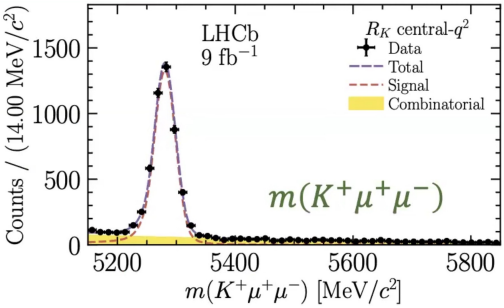
The Result: R_K Returns to 1.0

- Driving the denominator (N_{ee}) down mathematically forced the R_K ratio back up.
- **New R_K:** 0.994 ± 0.042 (Statistically consistent with 1.0)
- **New R_{K^*}:** 1.027 ± 0.072
- **The Verdict:** The 3σ tension completely evaporated. Lepton Flavour Universality holds perfectly within current LHCb precision.


LHCb
Data 9 fb^{-1}
Total fit
$B^+ \to K^+e^+e^-$
$B^+ \to J/\psi(e^+e^-)K^+$
Part. reco.
Combinatorial
Candidates per 24 MeV c^{-2}
$m(K^+e^+e^-)$ (MeV c^{-2})


LHCb
Data 9 fb^{-1}
Total fit
$B^+ \to K^+\mu^+\mu^-$
Combinatorial
Candidates per 7 MeV c^{-2}
$m(K^+\mu^+\mu^-)$ (MeV c^{-2})


LHCb
9 fb^{-1}
R_K central-q^2
Data
Total
Signal
Combinatorial
Misidentification
Partially reconstructed
$B^+ \to K^+J/\psi(\to e^+e^-)$
Counts / (32 MeV/c^2)
$m(K^+e^+e^-)$ [MeV/c^2]


LHCb
9 fb^{-1}
R_K central-q^2
Data
Total
Signal
Combinatorial
$m(K^+\mu^+\mu^-)$
Counts / (14.00 MeV/c^2)
$m(K^+\mu^+\mu^-)$ [MeV/c^2]

Summary: The State of Flavour Anomalies

1. Neutral Currents ($b \to s\ell^+\ell^-$): LFU?

- **LFU Restored:** The R_K and R_{K^*} anomalies vanished upon correcting hadronic misidentification in the electron channels.
- **The Muon Deficit:** The P_5' angular anomaly ($\Delta\mathcal{C}_9 \approx -1$) persists, but confusion with charm loop effects exist.

2. Charged Currents ($b \to c\tau\nu$) crisis

- **The Tau Excess:** A combined $\sim 3.2\sigma$ tension remains in the tree-level R_D and R_{D^*} ratios.
- **Theoretical Tension:** Explaining a tree-level anomaly requires strong/light New Physics (e.g., Leptoquarks), creating severe tension with the lack of direct particle discoveries at ATLAS and CMS.

3. The Path Forward

- The resolution of the R_K anomaly serves as a stark reminder of the complexity of hadronic backgrounds.
- Run 3 data from LHCb and independent cross-checks from Belle II are actively required to determine if the $R_{D^{(*)}}$ anomalies are genuine New Physics or stubborn systematic artifacts.